Smart Library Request Workflow in ServiceNow

Technical Documentation — Project Development Phase

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Academic & Technical Project Documentation

1. Introduction

The Project Development Phase represents the stage at which the requirements, plans, and design decisions established in all preceding phases are translated into a fully operational system. Where earlier phases were concerned with understanding what the system must do, who it must serve, and how its delivery should be organised, the development phase is concerned exclusively with execution — configuring, building, testing, and validating each system component within the ServiceNow platform until it meets the acceptance criteria defined during planning.

For the Smart Library Request Workflow, development was conducted entirely within a ServiceNow Personal Developer Instance, utilising the platform's native capabilities to avoid unnecessary custom code and to ensure long-term maintainability. The development activities followed the nine-milestone sequence established in the Project Planning Phase, beginning with foundational role configuration and progressing systematically through table development, field and reference configuration, workflow automation, user interface policies, access controls, and reporting. Each component was tested and validated before the next was introduced, ensuring that dependencies were respected and errors were caught early.

This document provides a detailed technical account of each development activity undertaken. It is intended to serve both as a record of what was built and how, and as a reference guide for administrators and support personnel responsible for maintaining the system following deployment. The sections that follow address each major development area in turn, describing the components created, the configuration decisions made, and the functional outcomes achieved.

2. Role Configuration

Role configuration was the first development activity undertaken, as all subsequent access control and permission decisions depend upon the existence of clearly defined roles. Two custom application roles were created within the scoped ServiceNow application, each corresponding to one of the primary user groups identified during the Requirement Analysis Phase. Establishing these roles at the outset of development ensured that every subsequent configuration artefact — including Access Control Lists, UI Policies, and report visibility settings — could be associated with the correct role from the moment of creation.

Student Role (lib_student)

The lib_student role is assigned to all registered library users who interact with the system as borrowers. Users holding this role interact primarily with the Borrow Request table, submitting new requests for available titles and monitoring the status of their existing requests through the self-service portal. The role is intentionally constrained to prevent any form of inventory management or administrative action.

Permissions granted:

Create new Borrow Request records; read their own Borrow Request records; read available Book records for selection purposes. No write, update, or delete access is granted to Book records, and no access is granted to records belonging to other users.

Librarian Role (lib_librarian)

The lib_librarian role is assigned to library staff who are responsible for the full operational management of the library system. Users holding this role have comprehensive access to both custom tables and are the designated approvers within the automated borrowing workflow. Librarians are responsible for maintaining the accuracy of the book catalogue, processing approval decisions, confirming book returns, and generating operational reports.

Permissions granted:

Full create, read, write, and delete access to all Book records; full create, read, write, and delete access to all Borrow Request records regardless of the submitting user; access to all pre-defined reports and the ServiceNow report builder; ability to take approval and rejection actions within the Flow Designer-driven workflow.

The separation of privileges enforced through these two roles is fundamental to the integrity and security of the system. It ensures that students cannot modify the book catalogue, access one another's request records, or bypass the approval process. Equally, it ensures that librarians retain the administrative visibility necessary to manage the collection and process requests efficiently. Both roles were tested in isolation using dedicated user accounts before any Access Control Lists were applied, confirming that role assignment alone produced the expected baseline behaviour.

3. Table Development

Following the completion of role configuration, two custom tables were created within the scoped ServiceNow application. These tables form the core data layer of the Smart Library Request Workflow, and all other system components — workflows, access controls, reports, and UI policies — interact with one or both of them. Both tables were created using ServiceNow's Studio application development environment and were assigned table names prefixed with the application scope identifier to maintain clean separation from platform system tables.

3.1 Book Table (u_book)

The Book table serves as the authoritative record for the library's physical collection. Each row in the table represents a single title held by the library and stores all metadata necessary to support catalogue management, availability tracking, and reporting. The table was extended from the base Task table to inherit standard platform fields such as the system number, creation date, and modified date, while custom fields were added to capture book-specific attributes. The u_status field is particularly significant, as its value governs both the reference qualifier applied to the Borrow Request form and the automated status transitions triggered by the approval workflow.

> **Table Purpose**
> The Book table (u_book) acts as the master inventory record for the library. Its availability status is the single source of truth for whether a title can be requested, and all status transitions are managed automatically by the system workflow to eliminate the risk of manual update errors.

3.2 Borrow Request Table (u_borrow_request)

The Borrow Request table records every borrowing transaction initiated within the system. Each row represents a single request submitted by a student and captures the complete lifecycle of that transaction — from initial submission through librarian decision to final return confirmation. The table maintains a reference relationship with the Book table through the u_book field, linking each request to the specific title being requested. This relationship is enforced at the data layer and is further constrained at the form level through the reference qualifier described in Section 5.

The relationship between the two tables is central to the system's operational logic. A Borrow Request record cannot exist without a valid reference to a Book record, and the approval workflow relies on this relationship to locate and update the correct Book record when a request is approved or a return is confirmed. This parent-child relationship also enables the Most Borrowed Books report, which aggregates approved requests by their associated book title to produce a ranked utilisation summary.

4. Field Configuration

Once both tables were created, fields were added to capture all data elements identified in the Data Requirements section of the Requirement Analysis Phase. Field types, labels, default values, and mandatory indicators were configured for each field to ensure data integrity and to support the downstream system logic that depends on field values — including the reference qualifier, workflow conditions, and report filters.

4.1 Book Table Fields

Field Name	Field Type	Description and Configuration Notes
u_title	String (255)	The full title of the book. Configured as a mandatory field with a unique constraint to prevent duplicate catalogue entries.
u_author	String (255)	The name of the primary author or editor. Mandatory field; free-text entry without controlled vocabulary.
u_isbn	String (20)	The International Standard Book Number for the edition. Validated against standard ISBN-10 and ISBN-13 formats using a field-level validation script.
u_category	Choice	Subject classification of the title. A controlled choice list was defined with values including Science, Engineering, Arts, Law, and General. Used as a filter dimension in reporting.
u_status	Choice	Current availability state of the book. Choice values: Available, Issued, Lost. Default value set to Available on record creation. This field drives the reference qualifier and is updated automatically by the workflow.
u_added_by	Reference (sys_user)	Auto-populated with the currently logged-in user upon record creation. Records which librarian added the title to the catalogue.
u_date_added	Date	Auto-populated with the current date upon record creation. Supports date-range filtering in inventory reports.

4.2 Borrow Request Table Fields

Field Name	Field Type	Description and Configuration Notes
u_requested_by	Reference (sys_user)	Auto-populated with the session user's sys_id upon record creation. Identifies the student who submitted the request. Read-only after submission.
u_book	Reference (u_book)	References the Book table. Subject to the reference qualifier described in Section 5, which restricts selection

		to records with a u_status value of Available. Mandatory field.
u_request_date	Date/Time	Auto-populated with the current date and time upon record creation. Read-only; cannot be modified after submission.
u_status	Choice	Lifecycle status of the borrow request. Choice values: Requested, Approved, Rejected, Returned. Default value set to Requested on creation. Updated automatically by the approval workflow.
u_approved_by	Reference (sys_user)	Populated by the workflow with the sys_id of the librarian who took the approval or rejection action. Read-only; not visible on the student-facing form.
u_decision_date	Date/Time	Auto-populated by the workflow at the moment the librarian takes an approval or rejection action.
u_expected_return	Date	Set by the librarian upon approval to specify the date by which the book must be returned. Enforced as mandatory by the UI Policy when the request status is Approved.
u_actual_return	Date	Recorded by the librarian when the physical book is returned. Triggers the workflow action that updates the Book record status back to Available.
u_rejection_reason	String (500)	A brief explanation provided by the librarian when rejecting a request. Included in the automated rejection notification sent to the student.

The field configurations across both tables were designed to minimise the data entry burden on users while maximising the accuracy and completeness of the records produced. Auto-populated fields eliminate transcription errors in key reference data such as the requesting user and submission timestamp. Controlled choice lists on status fields prevent the introduction of invalid values that could disrupt workflow logic or produce inaccurate report results.

5. Reference Qualifier Configuration

A reference qualifier was applied to the u_book field on the Borrow Request table to enforce a critical business rule: students may only submit a borrow request for a book whose current status is Available. Without this constraint, it would be possible for a student to request a title that is already issued to another borrower, creating conflicting records and undermining the integrity of the inventory tracking system.

The reference qualifier was implemented as a condition filter on the u_book reference field, restricting the lookup results displayed to the student when selecting a book on the Borrow Request form. The condition evaluates the u_status field on the Book table and returns only those records where the value equals Available. Records with a status of Issued or Lost are excluded from the selectable list entirely, meaning they are invisible to the student during the request submission process.

Reference Qualifier Condition

javascript:"u_status=Available" — Applied to the u_book reference field on the u_borrow_request table. This condition is evaluated dynamically at form load, ensuring that the available book list reflects the current inventory state at the moment of submission rather than a cached result.

The reference qualifier was validated using a test dataset containing Book records in each of the three possible status states. Test submissions confirmed that only Available records appeared in the selection list, and that the form correctly refreshed its available options when a book's status was changed between test runs. This validation was conducted using both the Student and Librarian test accounts to confirm consistent behaviour across roles.

6. Workflow Automation Using Flow Designer

The approval workflow represents the most consequential component of the Smart Library Request Workflow system. It governs the entire borrowing lifecycle — from the moment a student submits a request to the confirmation of a book's return — and replaces all manual notification and status update processes that would otherwise be required. The workflow was built using ServiceNow Flow Designer, which provides a graphical, low-code authoring environment that produces maintainable, versionable automation logic without requiring direct scripting.

The workflow is scoped within the library application and is activated as a published flow. It was developed iteratively, with each step tested individually before the full end-to-end path was validated. The complete workflow logic is described below.

01

Trigger — Record Created or Updated

The flow is triggered when a record is inserted into the Borrow Request table with a u_status value of Requested, or when an existing record's u_status field is updated to a value that initiates a new processing path. Using a combined trigger ensures the workflow responds correctly to both new submissions and return confirmations.

02

Action — Send Acknowledgement Notification to Student

Immediately upon trigger, the workflow dispatches an email notification to the student identified in the u_requested_by field. The notification confirms that the request has been received,

provides the book title and submission date, and advises that the student will be notified once the librarian has reviewed the request.

03

Action — Create Approval Task for Librarian

The workflow generates a structured approval task and assigns it to all users holding the lib_librarian role. The task is surfaced in the librarian's ServiceNow task list and includes the full details of the borrow request. The librarian is presented with clearly labelled Approve and Reject action options.

04

Decision — Evaluate Librarian Response

The workflow branches based on the librarian's action. If the librarian approves the request, execution proceeds to Steps 05, 06, and 07. If the librarian rejects the request, execution proceeds to Step 08. The workflow waits in an active state until one of these actions is taken, and the approval task remains open until a decision is recorded.

05

Action [Approval Path] — Update Book Status to Issued

Upon approval, the workflow updates the u_status field on the referenced Book record from Available to Issued. This update occurs automatically and immediately, ensuring that the book is no longer presented as available to other students submitting new requests. The u_expected_return date provided by the librarian is simultaneously written to the Borrow Request record.

06

Action [Approval Path] — Update Borrow Request Status to Approved

The u_status field on the Borrow Request record is updated from Requested to Approved, and the u_approved_by and u_decision_date fields are populated with the librarian's user reference and the current timestamp respectively.

07

Action [Approval Path] — Send Approval Notification to Student

An email notification is dispatched to the student confirming that their request has been approved. The notification includes the book title, the approval date, and the expected return date, providing the student with all information necessary to collect the book from the library.

08

Action [Rejection Path] — Send Rejection Notification to Student

If the librarian rejects the request, the workflow updates the Borrow Request status to Rejected and dispatches an email notification to the student. The notification includes the rejection reason recorded by the librarian, ensuring the student understands why the request was declined and can take appropriate action.

09

Action [Return Path] — Update Book Status to Available

When the librarian confirms the physical return of a book by populating the u_actual_return field on the Borrow Request record, the workflow detects this update and automatically changes the Book record status from Issued back to Available. The Borrow Request record status is simultaneously updated to Returned, completing the lifecycle of the transaction.

The implementation of this workflow eliminates all manual notification and status update tasks that would otherwise fall to library staff. Every status change is recorded automatically with a precise timestamp, every stakeholder receives timely notification at each stage of the process, and the inventory record is always accurate. The workflow also provides a complete, auditable history of every decision taken — the approving librarian, the decision date, and the outcome are all captured on the Borrow Request record and can be reviewed at any time.

7. UI Policy Implementation

UI Policies were configured on both custom tables to improve the usability and accuracy of the forms presented to students and librarians. UI Policies in ServiceNow are client-side rules that evaluate conditions in real time and dynamically adjust the form — making fields mandatory, read-only, or hidden based on the current state of the record. They do not replace server-side validation but complement it by providing immediate visual feedback to the user and preventing common data entry errors before submission.

7.1 UI Policies on the Borrow Request Form

UI Policy	Configuration and Purpose
Return Date Mandatory on Approval	When the u_status field equals Approved, the u_expected_return field is made mandatory and highlighted. This ensures that the librarian cannot complete an approval action without specifying a return date, which is required for overdue tracking and student notification.
Lock Core Fields After Submission	When the u_status field is any value other than Requested (i.e., the request has progressed beyond initial submission), the u_book and u_requested_by fields are set to read-only. This prevents accidental modification of the core transaction reference after the workflow has been triggered.
Show Rejection Reason Field on Rejection	The u_rejection_reason field is hidden by default and is made visible only when the u_status field is set to Rejected. This keeps the form uncluttered during normal approval processing and ensures that the reason field is prominently displayed when it is contextually relevant.
Hide Administrative Fields from Students	Fields including u_approved_by, u_decision_date, and u_rejection_reason are hidden from users holding the lib_student role, as these fields are operational in nature and are not relevant to the student's self-service experience.

7.2 UI Policies on the Book Form

UI Policy	Configuration and Purpose
Lock Status Field for Students	The u_status field on the Book form is set to read-only for all users assigned the lib_student role. Students may view book details and availability status but cannot alter inventory records.
Highlight Available Status	When the u_status field equals Available, the field is rendered with a green visual indicator to provide immediate, at-a-glance confirmation of availability to both students browsing the catalogue and librarians reviewing inventory.

The UI Policy configurations collectively reduce the likelihood of invalid data being submitted, ensure that forms present only the fields relevant to the current user and context, and reinforce the role-based access model at the interface level. They were tested thoroughly using both the Student and Librarian test accounts to confirm that the correct policies were applied to each role and that dynamic field behaviour functioned correctly across all relevant status transitions.

8. Access Control Configuration

Access Control Lists were implemented to enforce the role-based permission model at the data layer. While UI Policies govern what users see on their forms, ACLs govern what operations users are permitted to perform on the underlying records, regardless of the interface through which they are accessed. This distinction is important: a well-designed ACL configuration ensures that permissions are enforced even in scenarios where a user attempts to interact with the system through means other than the standard form interface, such as through the ServiceNow REST API.

ACLs were defined for each combination of table, role, and operation type. The principle of least privilege was applied throughout: each role was granted only the permissions explicitly required to perform its defined functions, and no additional access was granted speculatively. The resulting permission matrix is presented below.

Table / Object	Role	Create	Read	Write	Delete
Book Table (u_book)	lib_student	—	✓	—	—
Book Table (u_book)	lib_librarian	✓	✓	✓	✓
Borrow Request (own records)	lib_student	✓	✓	—	—
Borrow Request (all records)	lib_librarian	✓	✓	✓	✓
Reports	lib_student	—	—	—	—

Reports	lib_librarian	✓	✓	✓	—

Each ACL was tested independently using the dedicated role test accounts. Verification included both positive testing — confirming that permitted operations could be performed successfully — and negative testing — confirming that unpermitted operations returned an appropriate access denial response. For example, the Student test account was used to attempt a direct update of a Book record's u_status field, and the ACL configuration was confirmed to prevent this action at the platform level, independently of the UI Policy that hides the field on the form.

9. Report Development

Six operational reports were developed using the ServiceNow Report Builder and configured to draw from live transactional data held in both custom tables. All reports were scoped within the library application and configured to be accessible to users with the lib_librarian role. Report data is refreshed in real time, ensuring that the information presented to library management always reflects the current state of the system.

Report Name	Description
Most Borrowed Books	A bar chart and ranked list displaying book titles ordered in descending sequence by the number of associated approved Borrow Request records. Provides library management with a clear view of collection demand and informs procurement decisions for additional copies of high-demand titles.
Book Availability Summary	A pie chart and summary table presenting the distribution of Book records across the three status values: Available, Issued, and Lost. Enables librarians to assess the proportion of the collection in active circulation and identify titles requiring administrative follow-up.
Borrow Request Status Report	A tabular report listing all Borrow Request records with their current status, requesting student, book title, submission date, and expected return date. Filtered views are available for Requested, Approved, Rejected, and Returned statuses, allowing librarians to focus on specific lifecycle stages.
Overdue Requests Report	A list report identifying all Borrow Request records with a status of Approved where the u_expected_return date has passed and the u_actual_return field remains empty. Sorted by overdue duration in descending order to prioritise follow-up actions.
Request Volume by Period	A line chart presenting the total number of Borrow Request records created within a user-specified date range, grouped by week or calendar month. Supports capacity planning and the identification of peak demand periods within the academic calendar.

Category Utilisation Report	A bar chart aggregating approved Borrow Request records by the category of the associated Book record. Identifies which subject areas attract the greatest student borrowing activity, informing collection development strategy and faculty liaison planning.

Each report was validated against the test dataset to confirm that it returned accurate results and that calculated aggregations — such as borrow counts and category totals — were arithmetically correct. Report access was verified using both test account types to confirm that the lib_student role could not access the reporting interface, while the lib_librarian role had full access to all configured reports. Scheduling options for automated report distribution were confirmed to be available within the platform's native report delivery settings.

10. Development Outcome

The Project Development Phase successfully delivered a fully operational, role-based Library Management System within the ServiceNow platform. Each of the nine milestones defined in the Project Planning Phase was completed, validated, and signed off, and all functional requirements documented in the Requirement Analysis Phase have been implemented and confirmed through testing.

The system now provides students with a clean, intuitive self-service borrowing experience. Available books are clearly presented through a reference-qualified selection interface, request submissions are acknowledged automatically, and students receive timely email notifications at each stage of the approval process. The Student role's access boundaries ensure that each user's experience is appropriately scoped to their own transactions, with no visibility into other users' records or administrative functions.

Librarians have access to a centralised, workflow-driven operational interface that handles request routing, status management, and stakeholder communication automatically. The elimination of manual notification and status update tasks frees library staff to focus on higher-value activities, and the real-time availability of six pre-configured reports ensures that management decisions can be made on the basis of accurate, current data rather than manual tallies or periodic audits.

From a platform architecture perspective, the system has been delivered entirely within a scoped ServiceNow application using native platform features. No unsupported customisations or external integrations have been introduced. All configuration artefacts — tables, fields, workflows, ACLs, UI Policies, and reports — are documented, version-controlled, and encapsulated within the application scope, ensuring that the system can be maintained, extended, or transferred to a new support team without requiring reverse engineering of undocumented components.

In summary, the development phase has produced a system that is functionally complete, technically sound, and operationally ready. The Smart Library Request Workflow fulfils every objective established during the project planning phase and delivers the efficiency improvements, process

automation, and reporting capability that were identified as institutional priorities from the outset of the project.